Reg. No. | | | | | | | | | | |

[14ME55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2018

DESIGN OF MACHINE ELEMENTS - II

Time:3 Hours

Maximum Marks:70

Instructions:

i. Answer any **FIVE** full Questions choosing atleast two from each part.

ii. Missing data may be suitably assumed.

iii. Use of data hand book is permitted.

PART - A

1. Determine the maximum tensile, compressive and shear stress induced in a C frame of a hydraulic portable riveter shown in Fig.1. All dimensions are in mm. 14

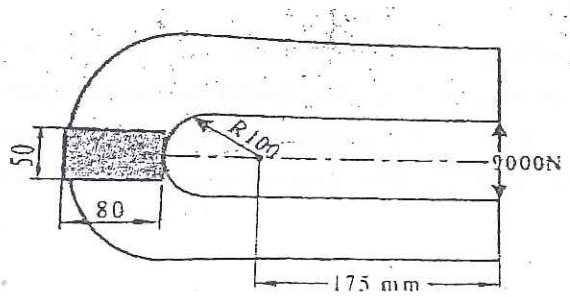

Fig.1

2. Design helical compression spring to support an axial load of 3000N The deflection under load is limited to 60mm. The spring index is 6.The spring is made of Chrome-vanadium steel and factor of safety is equal to 2. 14

3. Design a belt drive to transmit 25kW from a motor shaft rotating at 1500 rpm to a compressor running at 500 rpm. The motor pulley is 96mm in effective diameter and the center distance between the shafts is 1.5m 14

4. A forged steel pinion (SAE 1040) rotating at 400rpm drives a high grade cast iron gear. The transmission ratio is 4:1. The pinion has standard 20^0 full depth involute teeth of 4mm module. The face width of both gears is 40mm. How much power can be transmitted from the stand point of strength? 14

PART - B

5. A pair of bevel gears is to connect two shafts intersecting at 60^0. The gears are alloy steel of case hardened and precision cut with form cutters. The gear ratio is 5:1 the power transmitted is 30kW at 900rpm of the pinion. The teeth are 20^0 full depth. The pinion has 24 teeth. Determine the allowable stress, induced stress, beam strength of the weaker member. 14

6. Determine the dimensions of worm and worm wheel of a worm gear drive to transmit a power of 2kW at 1000rpm. The speed ratio is 20 and the center distance is 200mm. 14

7. A multiple plate clutch has steel on bronze is to transmit 8kW at 1440 rpm. The inner diameter of the contact is 80mm and the outer diameter of contact is 140mm. The clutch operates in oil with expected coefficient of friction of 0.1, the average allowable pressure is 0.35MPa. Assume uniform wear theory and determine the followings: 14

i. Number of steel and bronze plates

ii. Axial force required

iii. Actual maximum pressure

8. A lightly loaded journal bearing has a load of 1kN. The oil used is SAE60 and mean effective temperature of operates at 40^0C. The journal has a diameter of 50mm and the bearing has a diameter of 50.5mm. The speed of journal is 15000rpm. The L/d ratio is limited to 1.2 Determine the coefficient of friction and power loss due to friction. 14